

Roll No.

--	--	--	--	--	--	--	--

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: **B.Tech- CSE**Semester: **III**Name of the Course: **Logic Design & Computer Organization**Course Code: **TCS 308**Time: **3 Hours**Maximum Marks: **100**

Note:-

- (i) **All questions are compulsory.**
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each sub-question carries 10 marks.

Q1	(20marks)	
(a)	Implement full adder using 4:1 MUX when BC are selected lines.	CO1
(b)	What do you mean by essential prime implicants? Minimize the following Boolean expression using tabular method. $Y(A,B,C,D) = \sum m(5,7,8,10,13,15) + \sum d(0,1,2,3)$	
(c)	Implement 1 bit magnitude comparator using suitable decoder.	
Q2	(20 marks)	
(a)	Determine the characteristic equation of SR flip flop. Convert T flip flop into SR flip-flop.	CO2, CO3
(b)	Draw and explain the circuit of universal shift register.	
(c)	Draw and explain the circuit of BCD asynchronous counter.	
Q3	(20 marks)	
(a)	Draw and explain the operation of 4 bit twisted ring counter. Also determine total number of unused states.	CO2, CO3
(b)	Design a synchronous counter (using T flip flop) for a count sequence of 1, 3, 5, 7, 1.	
(c)	A clocked synchronous sequential circuit is described by two D flip flops, A and B; one input x and one output Z. If x=0, it works as up counter and if x=1, it works as down counter and when it reaches to minimum value output Z becomes 1 otherwise 0. (a) Draw state table and state diagram. (b) Draw the logic diagram of the circuit.	
Q4	(20 marks)	
(a)	A program runs in 500 micro second on a CPU with a clock rate of 2.8 GHz and CPI =2. If the CPI is reduced to 1.8, what would be the new execution time of CPU?	CO4, CO6

(b)	X=01110 and Y=11001 are two 5 bit binary numbers represented in 2's complement format. $Z = X - 3Y$. What will be value of Z in 8 bits format using 2's complement method?	
(c)	What do you mean by pipelining? Explain instruction pipelining with suitable example.	
Q5	(20 marks)	
(a)	Describe memory hierarchy based on parameters like speed, size, and cost and frequency.	CO5, CO6
(b)	Suppose access time of cache is 5 ns and cache hit rate is 80%, access time of main memory is 100 ns and main memory hit rate is 90%, access time of virtual memory is 10 μ s. What will be the average access time of memory hierarchy?	
(c)	Explain Amdahl's law and Flynn's taxonomy.	